Commercial Monitor: A Reproducible Multi-Source Framework for Research-to-Technology-to-Commercialization Forecasting

ZMath01 Research Project

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Abstract

We present Commercial Monitor, a reproducible framework for studying whether scientific activity is followed by measurable technology adoption and commercialization-related outcomes. The design separates research emergence from downstream outcomes rather than treating publication growth as a proxy for commercial success. The primary taxonomy contains five stable commercial-technology domains: Computer Systems & Engineering; AI & Applied Mathematics; Quantitative Finance; FinTech & Digital Finance; and Computational Quantum & Matter. Academic source categories are secondary mappings. ML, DL, foundation models, LLMs, inference, scaling, and AI agents are represented as subdomains under AI & Applied Mathematics because they overlap in abstraction level; scaling is treated as a cross-cutting property rather than a top-level domain. The runnable baseline combines publication signals with engineering and developer activity from public APIs, while adapters are planned for patent, funding, labor, company, and market datasets. Forecasts are evaluated with expanding-window temporal splits, probabilistic calibration, simple baselines, source-coverage diagnostics, and null experiments. Network measures such as effective conductance are retained as transparent graph baselines rather than assumed to measure commercialization. LLM assistance is constrained to semantic annotation and reporting; schemas, targets, evaluation boundaries, and scientific decisions remain human-controlled. This paper specifies the methodology and software architecture; it does not assert empirical performance before the live workflow is executed.

1 Introduction

Technology forecasting can conflate scientific novelty, technical adoption, intellectual-property translation, and market success. These processes are related but occur on different clocks and are measured by different observables. A publication may be scientifically important without becoming a product; conversely, a commercially important technology may be poorly represented by publication volume.

Commercial Monitor treats commercialization as a set of downstream outcomes. The primary forecasting object is

$$P(Y_k(t+h) = 1 \mid X_t), \tag{1}$$

where Y_k is a specified future outcome, h is a fixed horizon, and X_t contains only information observable by time t. This formulation makes temporal leakage a first-class concern.

2 Research Questions

1. Do research-emergence signals precede measurable technology/adoption signals?

2. Which combinations of research, engineering, IP, labor, funding, and company signals improve out-of-time prediction?

3. Are graph-based emergence measures useful beyond simple growth and persistence baselines?

4. Do results survive source-specific coverage controls and null-model tests?

The project does not define a single commercial-value score and does not interpret model output as an investment-return forecast.

3 Taxonomy and Data

The core taxonomy contains five stable commercial-technology domains. Source-specific arXiv/OpenAlex/GitHub/Sta Overflow mappings are secondary tags beneath them. The system also records orthogonal artifact types (software, model, algorithm, data, hardware, material, manufacturing, service) and commercial pathways (open source, developer tool, API, cloud, enterprise software, financial strategy/infrastructure, patent, hardware, manufacturing, licensing).

The distinction is important: taxonomy is not identical to live data sources. The initial implementation obtains research activity from OpenAlex and developer/community activity from GitHub and Stack Overflow. An arXiv adapter and downstream outcome schemas are provided as extension points rather than fabricated data.

Table 1: Outcome hierarchy.

Layer	Observable outcome
Research	future publication/concept emergence
Technology	repository, package, model, benchmark or developer adoption
IP	science-to-patent linkage and patent-family activity
Capital	grants and financing activity where reproducible
Labor	job-posting demand
Companies	company/startup formation
Market	procurement, licensing, product adoption or revenue proxies

4 Signals

For a non-negative count series x_t, growth indicators include year-over-year change and multi-year CAGR. Z-scores provide within-field standardization. Burst detection follows the two-state Poisson formulation of Kleinberg. A composite baseline is

$$S_{f,t} = \frac{\sum_j w_j \tilde{x}_{f,t,j}}{\sum_j w_j}, \tag{2}$$

where $\tilde{x}$ is a within-field normalized signal. Thus the index measures position relative to a field's own history, not comparable absolute market size.

5 Networks

Two graph families are distinguished. A future extension constructs concept co-occurrence graphs from documents. The current lightweight baseline constructs a field graph from correlations between field-level growth profiles. It is therefore explicitly named a growth-correlation graph, not a topic-cooccurrence graph.

For comparison with network-science baselines, the code implements a two-hop effective-conductance quantity:

$$G_{ab} = \sum_{v \in N(a) \cap N(b)} \frac{w_{av} w_{vb}}{w_{av} + w_{vb}}. \tag{3}$$

For a full weighted graph, effective resistance is

$$R_{\text{eff}}(i, j) = L^{+}_{ii} + L^{+}_{jj} - 2L^{+}_{ij}, \tag{4}$$

with conductance $1/R_{\text{eff}}$. These are network descriptors, not commercialization measures.

6 Prediction and Evaluation

Features include research growth, technology growth, community activity, burst states, current signal level, slope, and graph features. Logistic regression is trained in an expanding window. For test year t, only observations with year $< t$ are eligible for training. We report ROC-AUC when both test classes are present, average precision, and Brier score.

Simple baselines include a training-set class prior and graph-only specifications. Null experiments permute labels within pre-defined temporal blocks. The design favors ex-ante temporal evaluation over random splits because commercialization signals can have long and asynchronous lags.

7 Science-to-Invention Translation

The principal planned extension is explicit patent-paper linkage. Patent citations to scientific literature provide an observable of science-to-invention knowledge flow, although they are not equivalent to commercialization. A future implementation will preserve patent legal status, family structure, citation lag, and paper metadata as separate variables and predict future linkage using only information available before the patent event.

8 Human-Controlled Agent Boundary

LLMs can reduce the cost of semantic normalization, entity resolution, taxonomy suggestions, and report drafting. They should not define targets after seeing outcomes, select a winning model after observing test results, delete observations, or decide statistical significance. Operationally,

$$\text{LLM} = \text{assistant} \neq \text{scientific authority}.$$

Human-defined schemas, deterministic transformations, evaluation splits, and published metrics remain authoritative.

9 Software Architecture and Reproducibility

The repository uses Python, pandas/numpy/scipy/scikit-learn, NetworkX, YAML, pytest, and GitHub Actions. The initial portable artifact is CSV; Parquet/DuckDB can be introduced when volume warrants them. A deterministic sample mode separates software correctness from network availability. The monthly workflow is manually runnable by the maintainer and records provenance before production use.

10 Limitations

The executable baseline now uses prefix-only normalization, endpoint burst states, and historical growth-correlation graphs. Its prespecified target is at least 20% growth in GitHub topic-query counts over two calendar years, with zero-origin and unavailable future outcomes censored. Training labels must have outcome years strictly before the test year. Logistic feature scaling is fitted within training folds, and matched persistence, global-rate, graph-only, and research/adoption ablations accompany calibration and within-year label permutations. Descriptive uncertainty uses circular moving blocks of annual folds. None of these diagnostics establishes commercialization validity. The historical window is 2008–2025, providing 90 field-year units. Preserved source-query-year measurements overlap and are not additional independent forecasting units. Current API snapshots can revise historical counts and topic assignments; consequently these are retrospective snapshot backtests, not historical source-vintage simulations. Open bibliographic data have field and coverage biases. GitHub and Stack Overflow disproportionately represent software and English-speaking developer communities; they are poor universal measures for experimental physics or materials science. OpenAlex topic assignment is imperfect. API revisions and rate limits affect reproducibility. Patent, funding, labor, company, procurement, and revenue data have different licensing constraints and temporal delays.

Association between research activity and downstream adoption does not establish causal commercialization impact. The framework is intended for transparent forecasting and evidence aggregation, not causal claims or investment advice.

11 Relationship to Prior Approaches

The project is related to scientific trend forecasting, technology forecasting, patent-based emerging-technology detection, and network-based concept emergence. Its distinction is the outcome layer: it asks whether research signals predict independently observed downstream technology and commercialization outcomes. Concept-network reproduction can therefore serve as a benchmark without being the core commercial metric.

12 Expected Empirical Tests

The first empirical release should report source coverage by field and year; out-of-time performance against persistence and graph-only baselines; calibration; ablations for research, technology, community and graph features; permutation/null results; sensitivity to taxonomy boundaries and source availability; and paper–patent lag distributions when patent data are available. No performance number is asserted here before those experiments are executed.

13 Conclusion

Commercial Monitor provides a KISS, reproducible architecture for moving from research-trend detection toward empirically testable commercialization forecasting. Its central commitment is separation: research from adoption, proxies from outcomes, graph descriptors from commercial claims, and model fitting from future information.

References

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